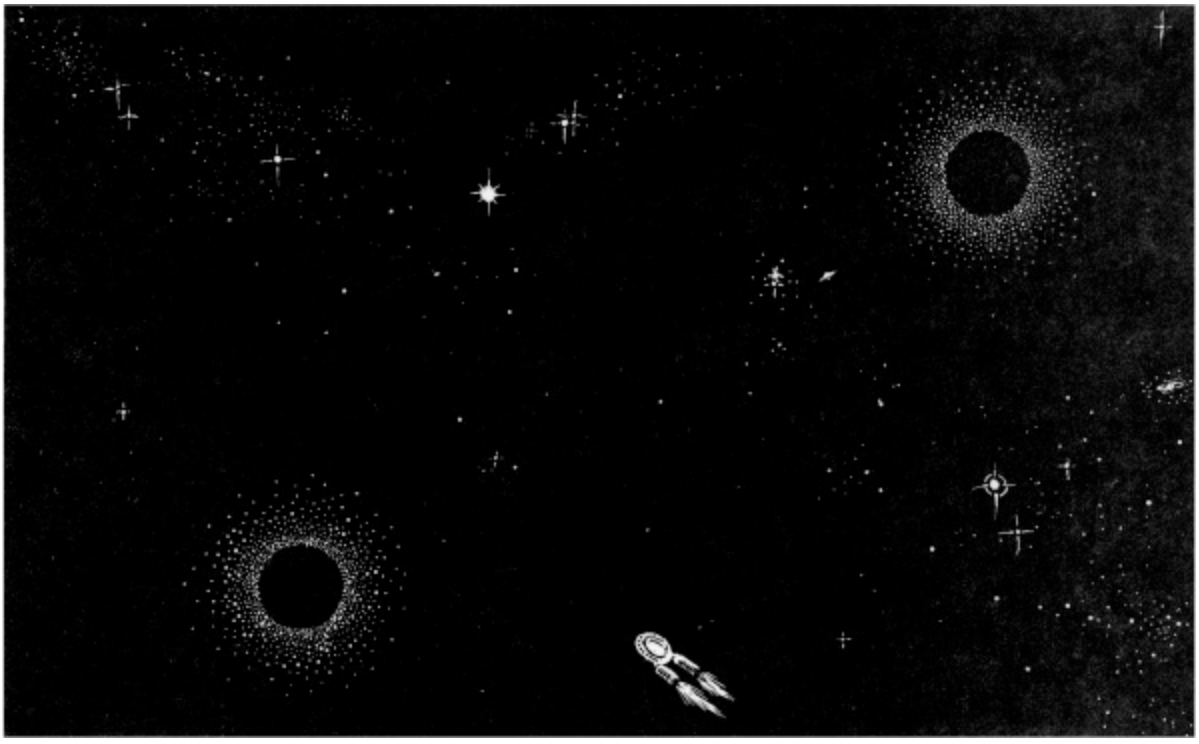
warp called a *wormhole*.[44] This space warp consists of two entrance holes (the wormhole's *mouths),* which look much like black holes but without horizons, and which can be far apart in the Universe (Figure P.7). Anything that enters one mouth finds itself in a very short tube (the wormhole's *throat)* that leads to and out of the other mouth. The tube cannot be seen from our Universe because it extends through *hyperspace* rather than through normal space. It might be possible for time to hook up through the wormhole in a different way than through our Universe, DAWN explains. By traversing the wormhole in one direction, say from the left mouth to the right, one might go backward in our Universe's time, while traversing in the opposite direction, from right to left, one would go forward. Such a wormhole would be a time warp, as well as a space warp.



P.7 The two mouths of a hypothetical wormhole. Enter either mouth, and you will emerge from the other, having traveled through a short tube (the worm-hole's throat) that extends not through our Universe, but through hyperspace.

The laws of quantum gravity demand that exceedingly tiny worm-holes of this type exist,[45] DAWN tells you. These quantum wormholes must be so tiny, just 10^{-33} centimeter in size, that their existence is only fleeting—far too brief, 10^{-43} second, to be usable for time travel. They must flash